Fischer J. (Moseley) Pettner

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Experience

Rivos, Inc. — *July 2024 - Present*
GPU Verification Engineer — *Fort Collins, CO*
- Lead DMA fixed-function unit verification for GPGPU-type accelerator, covering unit and subsystem-level testing.
- Built unit-level UVM testbench and wrote system-level assembly tests, drove ISA and RTL coverage to closure.
- Develop and maintain a regression orchestration tool that manages simulation workloads across a large HPC cluster (Slurm) and Cadence Palladium emulation hardware.
- Scaled to run over 50,000 tests nightly, executing 100 billion cycles across simulation and emulation combined.

NASA Jet Propulsion Laboratory — *June 2023 - June 2024*
FPGA Engineer II — *Pasadena, CA*
- Architected, implemented, and tested FPGA designs for a low Earth orbit optical interferometer/gravimeter mission, GRACE-C.
- Developed prototype RF hardware from requirements, using Altium/KiCad for schematic capture and board layout.
- Implemented designs for Xilinx/Microsemi FPGAs in Verilog and Python, and created CI pipelines for unit and system-level regressions with Questa.

The Art and Science of PCB Design - MIT PCB Design Class — *Nov 2022 - Feb 2023*
Lecturer — *Cambridge, MA*
- Created and taught a for-credit PCB design course at MIT, with over 100 enrolled students.
- Guided students through end-to-end design of a portable Bluetooth speaker, from requirements to bringup. Students completed schematic capture and board layout in Altium, performed SMD reflow assembly, and debugged their professionally-fabricated PCBs to create fully functional devices.
- Individually coached student projects, prepared and taught lectures, sourced materials, and acquired sponsors.

MIT Motorsports - Formula SAE — *Mar 2020 - Jun 2021*
Electrical Team Lead — *Cambridge, MA*
- Led the electrical team of MIT Motorsports, a student club that designs a Formula-style racecar for an annual competition.
- Architected the vehicle's electrical system, mentored team members, and maintained relationships with vendors and sponsors.
- Developed custom 20kW three-phase tractive inverter, modeling power and analog circuitry in LTSpice and designing PCBs in Altium. Implemented field-oriented control in C++ for embedded STM32 ARM Cortex-M7.

MIT STAR Laboratory - BeaverCube CubeSat — *Jun 2020 - Apr 2021*
Spacecraft Engineer — *Cambridge, MA*
- Developed a flight computer for the BeaverCube CubeSat mission, which provides data for hurricane prediction.
- Led firmware development for the embedded STM32 ARM Cortex-M4 housekeeping microcontroller in C++, and wrote embedded Linux software for the redundant Raspberry Pi Compute Modules onboard.
- Translated functional requirements into circuit design for a six-layer impedance-controlled PCB in Eagle.
- Deployed to low Earth orbit from the ISS in June 2021 via Falcon 9 launch and Nanoracks carrier.

SparkFun Electronics — *Jun 2019 - Aug 2019*
Experimental Product Engineer — *Niwot, CO*
- Created three new products for SparkFun's Qwiic product line, a set of I2C-enabled sensor modules.
- Designed circuits to specification, and used Eagle for schematic capture and board layout. Wrote open-source Arduino library in C++, and performed post-assembly hardware testing and quality assurance.

MIT Applied Mathematics Laboratory — *Aug 2018 - Aug 2021*
Experimental Physicist — *Cambridge, MA*
- Joint first author of *Active Topolectrical Circuits*, published in PNAS.
- Designed a network of analog nonlinear oscillators to demonstrate topological behavior similar to the Quantum Hall effect.
- Simulated circuit dynamics in MATLAB, designed a circuit using analog multipliers in LTSpice, and fabricated circuit boards designed in Eagle. Automated component characterization using PyVISA-controlled test equipment.

Education

Massachusetts Institute of Technology - Masters of Electrical Engineering and Computer Science — *June 2023*
GPA: 5.0/5.0 — *Cambridge, MA*
- **Thesis Title:** *Manta: An In-Situ Debugging Tool for Programmable Hardware.* Advised by Joseph Steinmeyer.
- Created an open-source, platform-agnostic tool for debugging and rapid prototyping of FPGA designs.
- Used CI to automate functional simulation, formal verification, and hardware-in-the-loop testing on Xilinx and Lattice FPGAs.
- Teaching Assistant for MIT's FPGA design class, and wrote new lab assignments for the course.
- **Coursework:** Power Electronics (Theory), Feedback and Control System Design, Underactuated Robotics, Robotic Manipulation, Quantum Measurement and Noise.

Massachusetts Institute of Technology - Bachelor's Degree in Electrical Engineering and Physics — *May 2022*
GPA: 4.6/5.0 — *Cambridge, MA*
- **EE Coursework:** Nanofabrication Project Lab, FPGA Project Lab, Power Electronics Lab, Mobile Autonomous Systems Lab, Signal Processing, Electromagnetics, Circuits and Electronics.
- **Physics Coursework:** Quantum Physics I & II, Quantum Computation, Intro Fusion Reactor Design, Advanced Electricity and Magnetism, Statistical Mechanics, Classical Mechanics I & II.

Skills

Hardware/EDA Tools	Altium, Eagle, KiCad, LTSpice, Vivado, Xcelium, Questa, Jasper Gold, Fusion 360
Software/Platforms	PyTorch, OpenCV, QuTip, Drake, Dagster, Docker, Kubernetes, Microsoft Azure
Languages	Python, (System)Verilog, MATLAB, C/C++, LaTeX, English (native), German (basic)